

Improving Reliability in Legal AI Through Multi-Agent Verification

Court Opinions Analyzer, Phase II — project status as of 5 October 2026

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1. Where the project stands

Capstone II tests whether a **multi-agent reading layer** catches incorrect answers on election-law opinions better than a **single model call** plus self-reported confidence — without replacing the human scholar as the final authority.

This week's snapshot. The full LEXIS corpus is now in a shared Postgres database, identified by citation (not file name), grouped into litigation families, and searchable with vector retrieval. The evaluation set is a frozen **50-decision** pilot. We have **not** yet run System A vs System B or written the Q1/Q2 answer key. That is the work of the next several weeks.

This updates the 23–24 September plan (then a 20-case Parquet file in the Capstone I app). Filenames such as *Craig v. Simon (2)* are **other decisions in the same lawsuit** (often district vs. appellate), not duplicates. Identity is the U.S. LEXIS citation. The corpus lives in shared Postgres (Supabase), not Capstone I SQLite. The Capstone I application remains the **System A baseline** (single Groq call, human-in-the-loop). It is not the corpus of record.

2. Research question (unchanged)

On the same frozen opinions and the same two questions, does **System B** (two independent readers + an adjudicator that preserves disagreement) outperform **System A** (one Groq call, Capstone I style) — especially on the temporal question Capstone I failed?

Question	Role in the experiment
Q1 — type of election	Control. Capstone I was already strong here.
Q2 — days between the trial ruling and the election	Known failure mode (temporal / Purcell-style reasoning).

Metrics remain accuracy vs. human labels, disagreement among agents, calibration, tokens, latency, and cost. Agents flag; validators and the scholar still decide. We are not building a chatbot for litigants.

3. Work completed since the September status report

Workstream	Status	What exists now
Word corpus ingest	COMPLETE	541 LEXIS .docx files parsed with rules (no LLM). 0 parse failures. 540 unique decisions loaded.
Case identity & families	COMPLETE	Primary key = LEXIS citation (86 files have no U.S. LEXIS line; keyed by source file). 427 litigation families; 383 prior/related links from the LEXIS header.
Chunking + retrieval	COMPLETE	14,987 chunks with character offsets (0 offset mismatches). Local embeddings; Postgres vector search. Smoke-tested on a Purcell-style query.
Frozen evaluation set	SELECTED	50 decisions, mid-length (about 2k–50k characters), at most 3 per family, election-related language. Pending human review of Q1/Q2 fitness.

Workstream	Status	What exists now
System A (Capstone I app)	AVAILABLE	Existing FastAPI / React HITL analyzer: one Groq call, validator, scholar review. Used as the single-agent baseline, not as the corpus store.
Q1/Q2 ground truth	NOT STARTED	Labels will be keyed by LEXIS citation (or filename if there is no cite). Human-authored.
System B (multi-agent)	NOT STARTED	Two readers + adjudicator on Groq (planned models: GPT-OSS 20B / 120B, Qwen 3.8 27B). Groq budget reserved for this, not for ingest.
A vs B evaluation run	NEXT	Cannot start until labels exist and System B can be called on the same 50 decisions.

4. How the pieces fit

Ingest is rule-based (no LLM). Embeddings are a small local model. Groq is reserved for the experiment. On the 50, a reader can still receive the **full opinion**. Retrieval is for the other ~490 longer opinions and for later evidence checking (an agent cites a character span; we verify it is actually in the opinion).

5. Team roles

Person	Owns	This period
Enock Katenda	Ingest, schema, retrieval	Corpus and retrieve are in place. Next: share the pipeline in the team repo.
Alex Ji	Models and agents (System B)	Next: two readers + adjudicator; log prompts, spans, confidence, cost.
Leandro Ardiles	Capstone I app, GitHub, Groq account	System A baseline stays. Next: official Groq access; merge ingest.
Tafadzwa Nyemba	Q1 / Q2 ground truth	50-decision list ready. Next: confirm or swap cases; write the answer key.

6. Next three weeks (planned)

- 1. This week (through 7 Oct, Week 2 checkpoint).** Put ingest/retrieve code on the shared GitHub repository. Invite the team to the shared database. Tafadzwa begins Q1/Q2 labels on the 50. Alex smoke-tests retrieve and drafts the reader/adjudicator interface. Dummy-AI path on System A remains available.
- 2. 8–21 Oct (Weeks 3–4).** Build System B behind the existing analysis call (do not rewrite the frontend first). Run Dummy mode, then Groq. Keep disagreement visible rather than collapsing to one answer.
- 3. Late October.** Run System A vs System B on the identical 50 decisions and the same two questions. Compare accuracy, disagreement, calibration, tokens, and cost. If retrieval work slips, the 50-case run can still use full opinion text. A temporal-reasoning pass and evidence checker (using stored character offsets) come only after that comparison is in hand.

7. Risks we are watching

- Labels are the gate.** We cannot score either system until Q1/Q2 ground truth exists for the frozen 50. Poor-fit cases get a new manifest version, not a silent regenerate.
- Nine of the 50 have no U.S. LEXIS citation** (often state-court exports). Those rows are keyed by source filename.
- Groq cost.** Ingest did not use Groq; agent runs will. We will log tokens and cost. We are not sending all 540 full opinions through an LLM.